

Infrastructure Monitoring with Zabbix

Server Engineering Labs

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Topics Covered:

- Infrastructure monitoring
- Zabbix server deployment
- Zabbix Agent2 configuration
- MariaDB integration
- Apache frontend configuration
- Linux service monitoring

Environment:

Debian Monitoring Server
Rocky Linux Client VM
Virtualized lab environment

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1. Introduction

This report documents the deployment and configuration of a monitoring infrastructure using Zabbix.

The laboratory focused on:

- Installing Zabbix Server
- Configuring MariaDB integration
- Deploying the Zabbix web frontend
- Configuring Zabbix Agent2
- Monitoring remote Linux hosts
- Testing alerts and service availability

The monitoring environment was deployed using Debian as the monitoring server and Rocky Linux as the monitored client system.

2. Environment Preparation

Before deploying Zabbix, network connectivity between machines was verified.

Useful commands included:

```
ping <host>  
ssh user@host
```

Firewall rules were also validated to allow monitoring traffic.

Required ports included:

- 10051/tcp — Zabbix Server
- 10050/tcp — Zabbix Agent
- 80/tcp — Apache Web Interface
- Custom SSH port for administration

```

[ubuntu@localhost ~]$ sudo firewall-cmd --permanent --add-port=10050/tcp
success
[ubuntu@localhost ~]$ sudo firewall-cmd --permanent --add-port=22022/tcp
Warning: ALREADY_ENABLED: 22022:tcp
success
[ubuntu@localhost ~]$ sudo firewall-cmd --permanent --add-port=80/tcp
success
[ubuntu@localhost ~]$ sudo firewall-cmd --reload
success
[ubuntu@localhost ~]$ sudo firewall-cmd --list-all
public (active)
  target: default
  icmp-block-inversion: no
  interfaces: enp0s3 enp0s8
  sources:
  services: cockpit dhcpv6-client http ssh
  ports: 22022/tcp 10050/tcp 80/tcp
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:

```

Figure 1: Firewall configuration allowing monitoring, SSH, and web service traffic required by the Zabbix infrastructure.

3. Installing Zabbix Server

The official Zabbix repository was added to the Debian system.

Example:

```

wget https://repo.zabbix.com/...
sudo dpkg -i zabbix-release.deb
sudo apt update

```

The required packages were then installed:

```

sudo apt install zabbix-server-mysql \
                zabbix-frontend-php \
                zabbix-apache-conf \
                zabbix-sql-scripts \
                zabbix-agent2

```

4. MariaDB Database Configuration

Zabbix requires a relational database backend.

A database and dedicated user were created:

```

CREATE DATABASE zabbix;
CREATE USER 'zabbix'@'localhost'
IDENTIFIED BY 'secure_password';

```

```
GRANT ALL PRIVILEGES ON zabbix.*
TO 'zabbix'@'localhost';

FLUSH PRIVILEGES;
```

The initial schema was imported:

```
zcat /usr/share/zabbix-sql-scripts/mysql/server.sql.gz \
| mysql -u zabbix -p zabbix
```

5. Zabbix Server Configuration

The Zabbix server configuration file was edited:

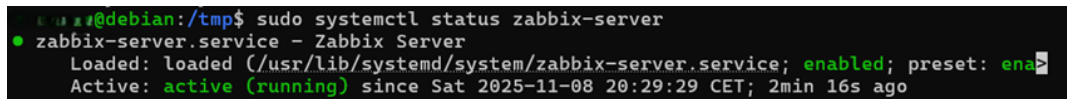
```
/etc/zabbix/zabbix_server.conf
```

The database password parameter was configured:

```
DBPassword=secure_password
```

The service was restarted afterward:

```
sudo systemctl restart zabbix-server
```



```
root@debian:/tmp$ sudo systemctl status zabbix-server
● zabbix-server.service - Zabbix Server
   Loaded: loaded (/usr/lib/systemd/system/zabbix-server.service; enabled; preset: ena
   Active: active (running) since Sat 2025-11-08 20:29:29 CET; 2min 16s ago
```

Figure 2: Zabbix server service successfully running after backend configuration and initialization.

6. Apache Frontend Configuration

The Apache frontend was configured to serve the Zabbix web interface.

Apache configuration was validated:

```
sudo systemctl status apache2
```

The PHP timezone was also configured:

```
date.timezone = Europe/Madrid
```

Apache was restarted after applying changes:

```
sudo systemctl restart apache2
```

6.1. UTF-8 Locale Troubleshooting

During deployment, locale issues affected the frontend checks.

The system locale was updated to use UTF-8 encoding:

```
sudo locale-gen en_US.UTF-8
sudo update-locale
```

Apache environment variables were adjusted accordingly.

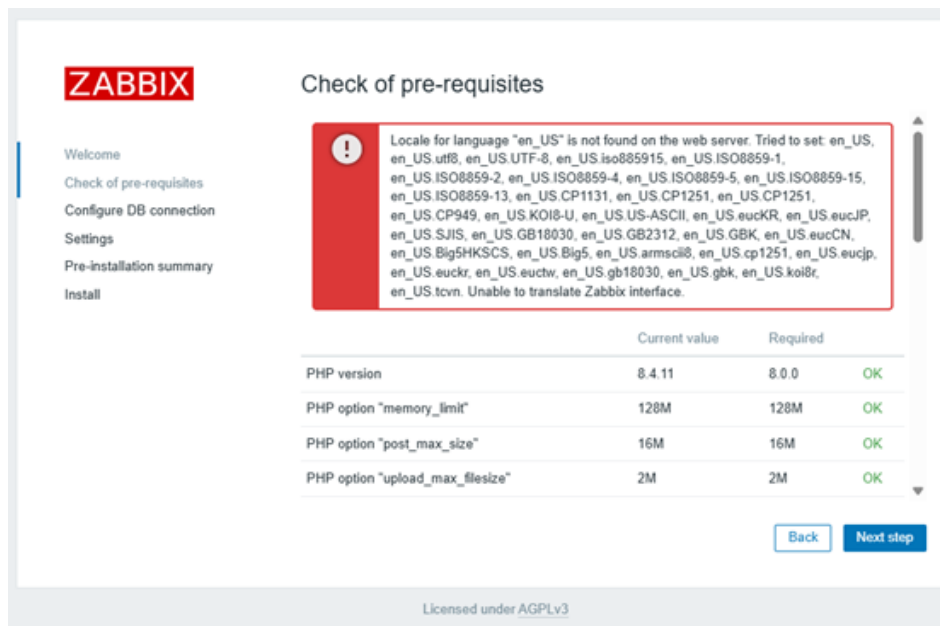


Figure 3: Zabbix frontend prerequisite validation highlighting locale configuration checks during deployment.

7. Installing and Configuring Zabbix Agent2

The monitored Rocky Linux machine was configured with Zabbix Agent2.

Installation:

```
sudo dnf install zabbix-agent2
```

The agent configuration file was edited:

```
/etc/zabbix/zabbix_agent2.conf
```

Important parameters:

```
Server=<zabbix_server_ip>
ServerActive=<zabbix_server_ip>
Hostname=rocky-monitor-node
```

The service was enabled and started:

```
sudo systemctl enable zabbix-agent2
sudo systemctl restart zabbix-agent2
```

8. Frontend Host Registration

The Rocky Linux machine was added as a monitored host in the Zabbix frontend.

The following information was configured:

- Hostname
- Agent IP address
- Monitoring templates
- Availability checks

Monitoring metrics became visible after successful registration.

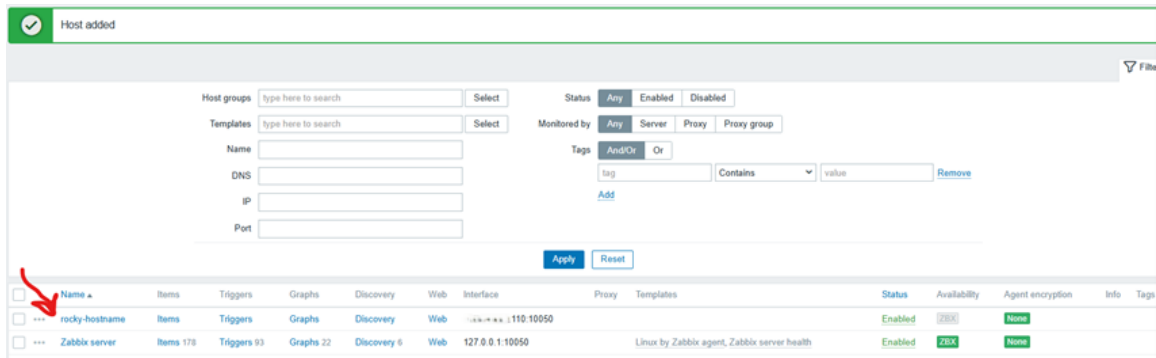


Figure 4: Successful registration of the monitored host in the Zabbix frontend interface.

9. Monitoring Validation

Several tests were performed to validate the monitoring infrastructure:

- Service availability checks
- Agent communication validation
- CPU and memory monitoring
- Trigger and alert simulation

The Zabbix dashboard successfully displayed system metrics from the monitored host.

10. Challenges Encountered

Several issues appeared during deployment:

- Database authentication problems
- Apache frontend configuration issues
- UTF-8 locale incompatibilities
- Firewall restrictions
- Agent communication failures

Troubleshooting these issues provided valuable practical experience in monitoring infrastructure administration.

11. Conclusion

This laboratory introduced the deployment of a complete infrastructure monitoring solution using Zabbix.

The exercises demonstrated:

- Monitoring server deployment
- Agent-based infrastructure monitoring
- Database integration

- Frontend configuration
- Linux service administration
- Monitoring validation and troubleshooting

These concepts are highly relevant in infrastructure engineering, DevOps, system administration, and production monitoring environments.